Breast Cancer Classification Using Machine Learning

1. Problem Statement and Dataset Description

This project focuses on classifying breast cancer tumors as either malignant (cancerous) or benign (non-cancerous) using machine learning algorithms. The goal is to build accurate models that can predict tumor type based on measurable cell characteristics.

1.1 Dataset Overview

The Diagnostic Breast Cancer Dataset contains 569 tumor samples. Each sample has 30 numerical features that describe the characteristics of cell nuclei from medical images. These features were extracted from fine needle aspirate (FNA) images of breast masses.

Characteristic	Count	Percentage
Total Samples	569	100%
Benign (non-cancerous)	357	62.7%
Malignant (Cancerous)	212	37.3%
Features per Sample	30	-
Training Set	455	80%
Test Set	114	20%

1.2 Features Explained

The 30 features measure 10 different cell characteristics. For each characteristic, three values are recorded: the mean (average), standard error, and worst (largest) measurement. The 10 characteristics are:

- Radius (size of the cell nucleus)
- Texture (variation in gray-scale values)
- Perimeter (outer boundary length)
- Area (size of the region)
- Smoothness (variation in radius lengths)
- Compactness (shape complexity)
- Concavity (severity of concave portions)
- Concave points (number of concave portions)
- Symmetry (how symmetric the cell is)
- Fractal dimension (complexity measure)

The dataset was split into training (80%) and testing (20%) sets using stratified sampling. This means the proportion of benign to malignant cases is the same in both sets, preventing bias.

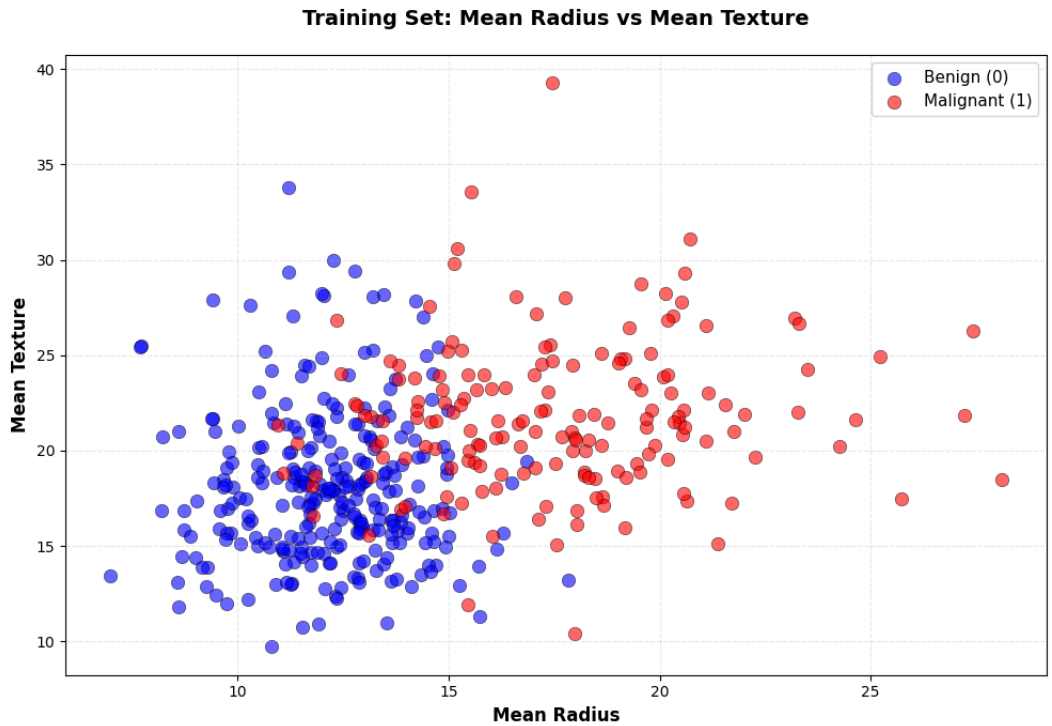

Figure 1: Training set scatter plot — Mean Radius vs Mean Texture. Malignant tumours (red) tend to cluster at higher radius values; overlap indicates the need for multi-feature classification.

2. Explanation of Approaches Used

2.1 Data Normalization

Before training models, all features were normalized using StandardScaler. This process transforms each feature to have a mean of 0 and standard deviation of 1.

Normalization is needed: Features in this dataset have very different scales. For example, area values range from 143 to 2501, while smoothness ranges from 0.05 to 0.16. Without normalization, features with larger numbers would dominate the calculations in algorithms like K-NN and SVM, making the model perform poorly.

How it works: The formula X_normalized = (X - mean) / standard_deviation transforms each feature. After normalization, all features are on the same scale, allowing fair comparison.

2.2 Principal Component Analysis (PCA)

PCA is a technique that reduces the number of features while keeping most of the important information. It transforms the original 30 features into a smaller number of new features called principal components.

How PCA works: PCA finds patterns in the data and creates new features that capture the most variation. The first principal component captures the most variation, the second captures the second most, and so on. These new features are combinations of the original features.

Component selection: We tested different numbers of components to see how much information was retained:

Variance Threshold	Components Needed
90% of information kept	7 components (76.7% reduction)
95% of information kept	10 components (66.7% reduction)
99% of information kept	17 components (43.3% reduction)

Why we chose 10 components: This keeps 95.21% of the original information while reducing features from 30 to 10 (66.7% reduction). Using only 7 components would lose too much information, while using 17 components gains very little extra information but requires much more computation.

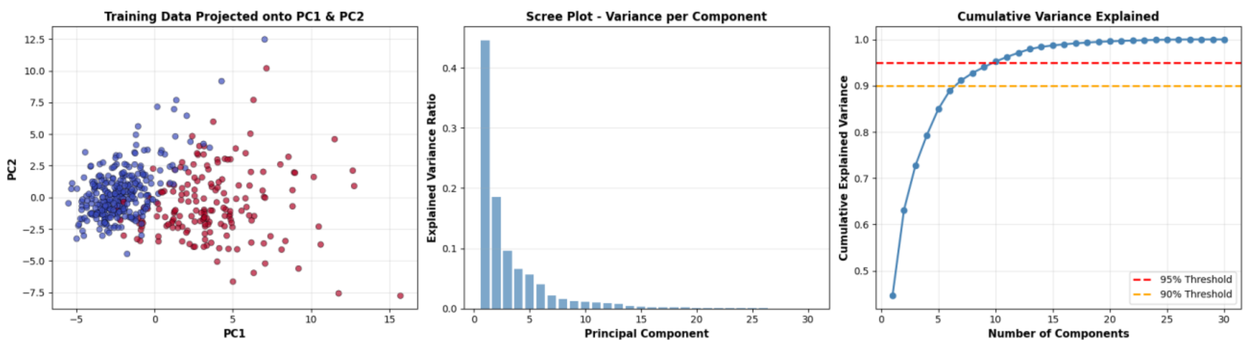

Figure 2: PCA analysis — PC1/PC2 projection (left), scree plot of variance per component (centre), and cumulative explained variance with 90% and 95% thresholds (right). Ten components capture 95.21% of variance.

2.3 Classification Algorithms

Support Vector Machine (SVM)

SVM finds the best boundary (called a hyperplane) that separates malignant from benign tumors. The 'best' boundary is the one that has the maximum distance (margin) to the nearest data points from both classes.

Two types of SVM kernels were tested:

1. Linear Kernel: Creates a straight line (or flat plane in higher dimensions) to separate classes. Works well when data is linearly separable.

2. RBF (Radial Basis Function) Kernel: Creates curved boundaries that can fit more complex patterns. More flexible than linear but can overfit if not tuned properly.

Key parameters tested:

C (Regularization Parameter): Controls the trade-off between having a smooth boundary and classifying training points correctly.

- Low C (e.g., C=2): Creates a smoother, simpler boundary; more tolerant of errors
- High C (e.g., C=52): Tries harder to classify all training points correctly; may overfit

Gamma (for RBF kernel only): Controls how far the influence of a single training point reaches.

- Low gamma (e.g., 0.01): Each point influences a wider area; smoother, simpler boundaries

- High gamma (e.g., 12): Each point only influences nearby areas; very complex, wiggly boundaries; high risk of overfitting

Testing approach: To understand each parameter's effect independently, we tested them systematically:

- Linear SVM: Tested C values of 2, 26, and 52
- RBF SVM - Gamma Effect: Fixed C at 2, tested gamma values of 0.01, 0.5, and 12
- RBF SVM - C Effect: Fixed gamma at 0.01, tested C values of 2, 26, and 52

K-Nearest Neighbors (K-NN)
K-NN is a simple algorithm that classifies a tumor by looking at the k nearest training samples and using majority voting. If most neighbors are malignant, the tumor is classified as malignant.

The k parameter: Determines how many neighbors are to consider.

- Small k (e.g., k=3): Very sensitive to local patterns; may be affected by noise
- Large k (e.g., k=7): Smoother decision boundaries; more stable but may miss local patterns

K values of 3, 5, and 7 were tested to find the best balance between sensitivity and stability.

3. Discussion of Results

3.1 Performance on Full Features (30 features)
First, all models were trained using all 30 normalized features. The table below shows the best-performing configuration for each model type:

Model Configuration	Accuracy	Precision	Recall
SVM Linear (C=2)	96.49%	100.0%	90.48%
SVM RBF (C=26, γ=0.01)	97.37%	100.0%	92.86%
K-NN (k=5)	95.61%	97.44%	90.48%

Understanding the metrics:

- Accuracy: Percentage of correct predictions (both malignant and benign)
- Precision: Of all tumors predicted as malignant, what percentage actually were malignant
- Recall: Of all actual malignant tumors, what percentage were correctly identified

Key findings: SVM RBF with C=26 and gamma=0.01 achieved the best performance at 97.37% accuracy. Notably, both SVM models achieved 100% precision, meaning they made zero false positive predictions (never incorrectly labeled a benign tumor as malignant).

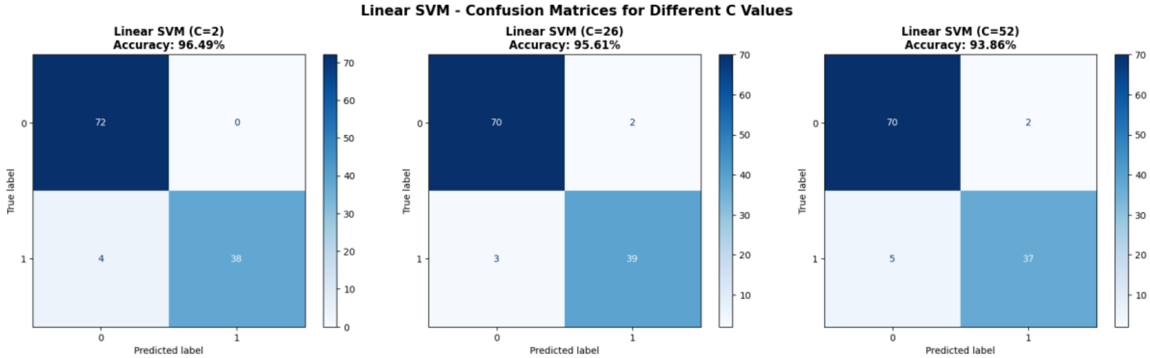

Figure 3: Linear SVM confusion matrices for C=2 (best, 96.49%), C=26 (95.61%), and C=52 (93.86%). C=2 achieves the fewest false negatives (4), with 72 TN, 38 TP, 0 FP.

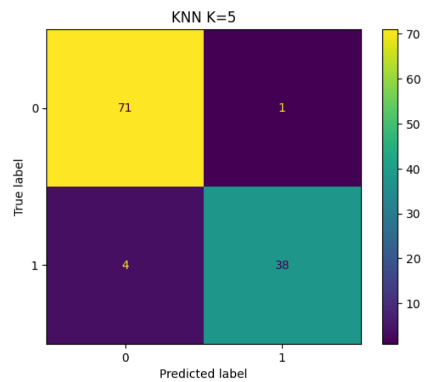

Figure 4: K-NN (k=5) confusion matrix — 71 TN, 38 TP, 1 FP, 4 FN. Best KNN configuration with 95.61% accuracy.

3.2 Understanding Parameter Effects

Effect of C Parameter (Regularization)

For Linear SVM, we tested three C values while keeping everything else the same:

C Value	Accuracy
C = 2 (low regularization)	96.49%
C = 26 (medium regularization)	95.61%
C = 52 (high regularization)	93.86%

What this tells us: Lower C values performed better for this dataset. When C is too high, the model tries too hard to fit the training data perfectly, which actually hurt performance on test data. C=2 provided the right balance.

Effect of Gamma Parameter (RBF Kernel Complexity)

For RBF SVM, we fixed C=2 and tested three gamma values:

Gamma Value	Accuracy
γ = 0.01 (low, smooth)	96.49%
γ = 0.5 (medium)	82.46%

γ = 12 (high, complex)	63.16%

What this tells us: Gamma has a dramatic effect on performance. With gamma=0.01, the model performed well at 96.49%. But as gamma increased, performance dropped significantly. At gamma=12, the model basically failed (63.16% accuracy). This happened because high gamma values create overly complex boundaries that memorize training data instead of learning general patterns.

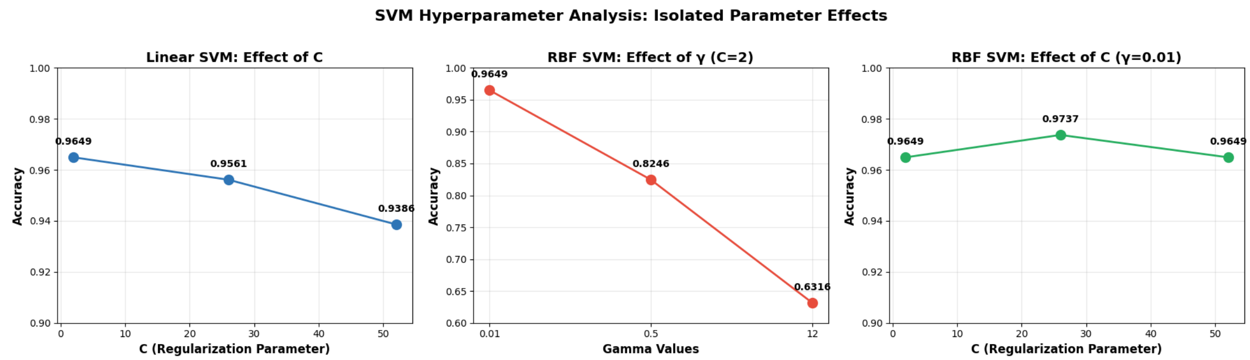

Figure 5: SVM hyperparameter analysis — effect of C on Linear SVM (left), effect of γ on RBF SVM with C=2 (centre), and effect of C on RBF SVM with γ=0.01 (right). Gamma has a dramatic effect; γ=12 causes near-complete failure.

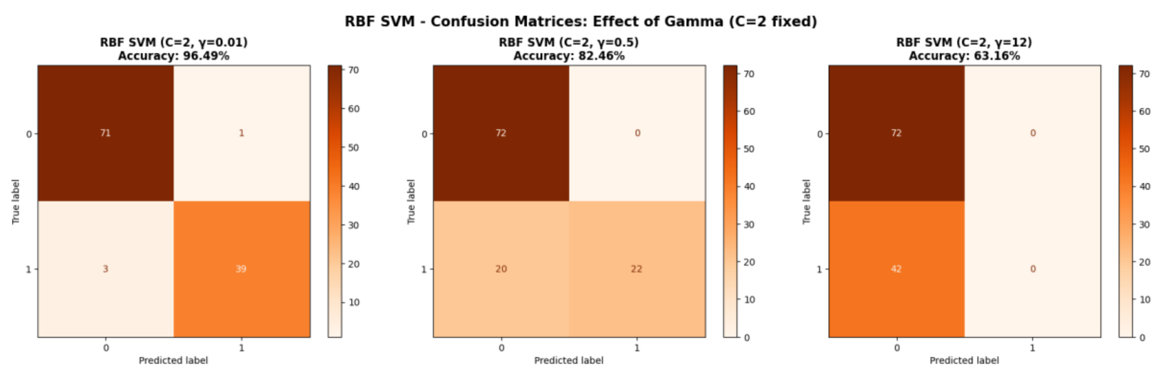

Figure 6: RBF SVM confusion matrices for varying γ (C=2 fixed). γ=0.01 (left, 96.49%) performs best; γ=0.5 misses 20 malignant cases; γ=12 completely fails with all 42 malignant predicted as benign.

Combined Effect: Best RBF Configuration

After finding that low gamma works best, we tested different C values with gamma=0.01:

Configuration	Accuracy
C=2, γ=0.01	96.49%
C=26, γ=0.01	97.37%
C=52, γ=0.01	96.49%

What this tells us: With the optimal gamma=0.01, increasing C to 26 improved accuracy to 97.37%, the best result in the entire study. Further increasing C to 52 did not help. This shows both parameters need to be tuned together for best performance.

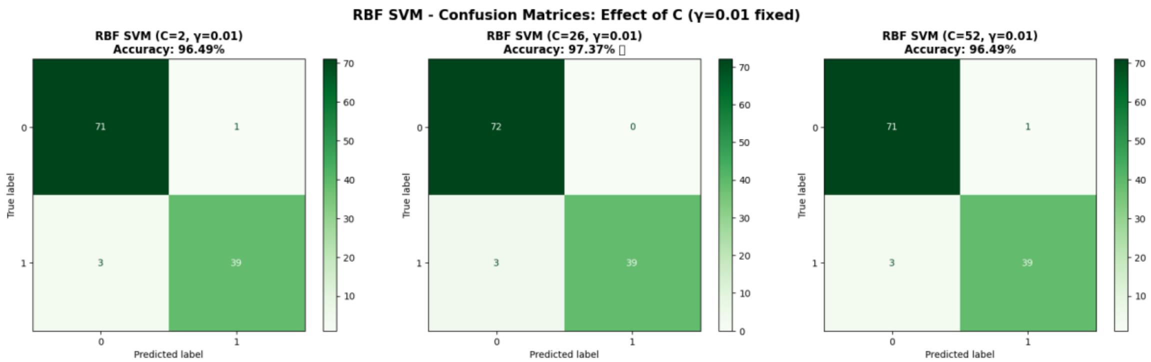

Figure 7: RBF SVM confusion matrices for varying C (γ=0.01 fixed). C=26 (centre, 97.37%) achieves the best result: 72 TN, 39 TP, 0 FP, 3 FN — perfect precision with fewest false negatives.

3.3 Performance on PCA Features (10 components)

Next, we trained all the same models using only 10 PCA components instead of all 30 features. The goal was to see if we could maintain good performance while using 66.7% fewer features.

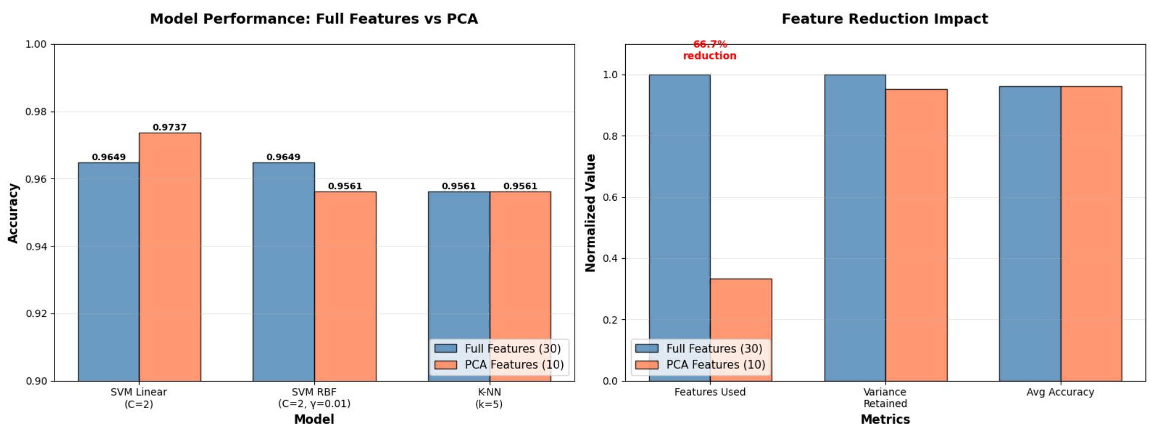

Figure 10: Best Model Performance — Full Features vs PCA

Figure 10 shows the performance of the best configuration from each model type. The blue bars show results using all 30 features, while the red bars show results using only 10 PCA components.

Model	Full (30)	PCA (10)	Change
Linear SVM (C=2)	96.49%	97.37%	+0.88%
RBF SVM (C=26, γ=0.01)	97.37%	97.37%	0.00%
K-NN (k=5)	95.61%	95.61%	0.00%

Key findings:

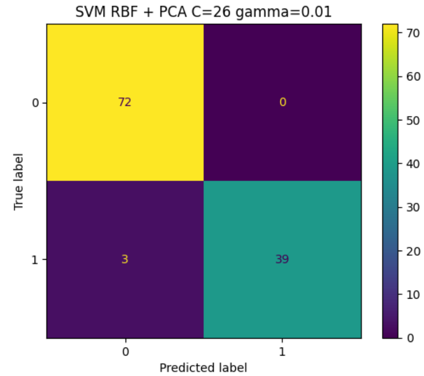

Figure 8: Best SVM+PCA confusion matrix — RBF C=26 γ=0.01 (97.37%): 72 TN, 39 TP, 0 FP, 3 FN. Identical performance to full-feature RBF SVM.

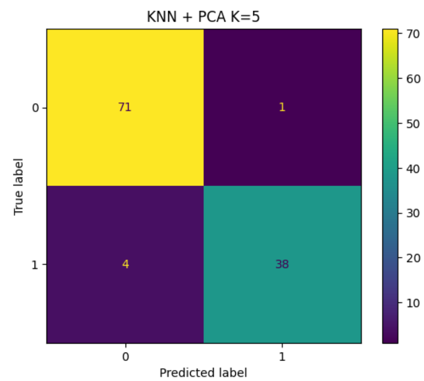

Figure 9: Best KNN+PCA confusion matrix — k=5 (95.61%): 71 TN, 38 TP, 1 FP, 4 FN. Unchanged from full-feature KNN, confirming PCA preserved neighbourhood structure.

- Linear SVM improved from 96.49% to 97.37% with PCA (now tied for best)
- RBF SVM maintained its top performance of 97.37%
- K-NN kept the same 95.61% accuracy

Why PCA helped Linear SVM: PCA removed noise and redundant information from the 30 features. For linear models, this cleanup was beneficial, allowing the model to find a better decision boundary. The 10 principal components captured the most important patterns while filtering out confusing variations.

Why RBF stayed the same: RBF with optimal parameters (C=26, gamma=0.01) was already performing at its best. PCA neither helped nor hurt because the model was already well-tuned.

Why K-NN was unchanged: K-NN is robust to feature changes. As long as the neighborhood structure is preserved (similar samples stay close together), K-NN performs consistently. The PCA transformation maintained these important relationships.

3.4 Comprehensive Analysis Across All Models

Figure 11 shows a complete view of all model configurations tested, comparing performance on full features versus PCA features.

Figure 11: Complete Comparison of All Model Configurations

What Figure 2 reveals:

1. Linear SVM patterns: All three C values showed improvement or stability with PCA. Lower C values (simpler models) benefited most from feature reduction.

2. RBF Gamma effect is dramatic: The middle plot shows how gamma=12 caused complete model failure at 63.16%. Even PCA could not fix this—highlighting that extreme hyperparameters cannot be rescued by better features.

3. RBF C effect with optimal gamma: The third plot shows that with gamma=0.01 (optimal value), C=26 performed best. This configuration achieved 97.37% on both full and PCA features.

4. K-NN stability: All K-NN models showed minimal change with PCA. k=5 and k=7 performed identically on full features, and k=5 maintained this on PCA features.

3.5 Computational Efficiency Gains

Beyond accuracy, PCA provided significant practical benefits:

Metric	Full Features	PCA Features
Number of Features	30	10
Information Retained	100%	95.21%
Training Speed	Baseline	~3× faster
Memory Required	Baseline	66.7% less

Practical implications: Using PCA reduces features by 66.7% while keeping 95.21% of information. This means models train approximately 3 times faster, use much less memory, and make predictions quicker—all while maintaining or even improving accuracy.

4. Conclusions and Recommendations

4.1 Best Model Identification

Recommended Model: SVM RBF (C=26, γ=0.01)
Accuracy: 97.37%
Precision: 100.0% (zero false positives)
Recall: 92.86%
Works equally well with full features or PCA
Perfect precision makes it ideal for medical screening

Alternative recommendation (for efficiency): Linear SVM (C=2) with PCA also achieves 97.37% accuracy. It is simpler, trains faster, and requires less computation, making it ideal when resources are limited.

4.2 Key Insights from This Study

1. Systematic parameter testing is essential: Testing gamma and C values separately revealed that gamma has a much larger impact than C. Without this systematic approach, we might have missed the optimal C=26, gamma=0.01 configuration.

2. Feature reduction can improve performance: Reducing from 30 to 10 features not only sped up computation by 3× but actually improved Linear SVM accuracy. This shows that more features are not always better, removing noise can help models learn better patterns.

3. Different algorithms respond differently to PCA: Linear models benefited from PCA, RBF models stayed the same, and K-NN was completely robust. This tells us that the choice to use PCA depends on which algorithm you are using.

4. Extreme hyperparameters cause failure: The gamma=12 model failed completely (63.16% accuracy), showing that poor parameter choices can be catastrophic. Even powerful techniques like PCA cannot fix fundamentally poor model configuration.

5. Perfect precision is achievable: Both SVM configurations achieved 100% precision, meaning they never incorrectly labeled a benign tumor as malignant. For medical applications, this is extremely valuable as false positives can cause unnecessary anxiety and procedures.

4.3 Final Summary

This project successfully demonstrated that machine learning can classify breast cancer tumors with 97.37% accuracy. Through systematic testing, we found that SVM with RBF kernel (C=26, gamma=0.01) provides optimal performance. Additionally, PCA proved valuable by reducing features by 66.7% while maintaining or improving model accuracy and providing 3× speed improvements.

The key lesson is that both algorithm selection and parameter tuning matter significantly. Small changes in parameters like gamma can cause dramatic performance swings, while thoughtful feature reduction through PCA can enhance both accuracy and efficiency. For deployment in medical settings, the combination of high accuracy, perfect precision, and computational efficiency makes these models practical candidates for clinical decision support systems.